

# popquiz momentum - review BONUS points ↕

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:43pm

## Quiz Instructions



### Question 1 12.5 pts

A freight train rolls along a track with considerable momentum. If it rolls at the same speed but has twice as much mass, its momentum is

hint: momentum is mass x velocity

☐

zero.

☐

four times as much.

☐

twice.

☐

unchanged.



## Question 2 12.5 pts

A same-size iron ball and wooden ball are dropped simultaneously from a tower and reach the ground at the same time. The iron ball has a greater

(its tricky. its a free-fall.  $V = \sqrt{2gh}$  from conservation of energy. Think. no 2nd attempt)

☐

acceleration.

☐

momentum.

☐

none of the above

☐

all of the above

☐

speed.



## Question 3 12.5 pts



Véronic is an asteroid that orbits the Sun between Jupiter and Mars. It's a bit smaller than our Moon. It's spherical.

Its radius is **476 km** and its mass about  **$9.1 \times 10^{20}$  kg**.

What is the escape speed for Veronic? (that is the speed that a spacecraft or an astronaut needs to escape gravity). (**slide 49 unit 7** use equation for  $V$ ).  $G = 6.6743 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$

Convert km to m.

Note : escape speed for Earth is about 11,000 m/s (about 7 miles/s)

☐

about 50 m/s (about 100 miles per second)

☐

about 5000 m/s (about 10,000 miles/s)



about 500m/s (about 1000miles/s)



about 5 m/s (about 10miles/s)



whaaaaat ?



#### Question 4 12.5 pts

When a boxer moves into an oncoming punch, the force experienced is



all of the above



no different, but the timing is different.



increased.



decreased.



#### Question 5 12.5 pts

To catch a fast-moving ball, you extend your hand forward before contact with the ball and let it ride backward in the direction of the ball's motion. Doing this reduces the force of contact on your hand principally because the



time of contact is decreased.



none of the above



relative velocity is less.



time of contact is increased.



force of contact is reduced.



#### Question 6 12.5 pts

Padded dashboards in cars are safer in an accident than non-padded ones because passengers hitting the dashboard encounter



increased momentum.



lengthened time of contact.



shorter time of contact.



decreased impulse.



### Question 7 12.5 pts

Which one experience the larger change in momentum ? (suppose the same mass and time if impact.) A bouncing ball or a tomatoe dropped from the same height. they have the same mass



I hate tomatoes



I missed the class



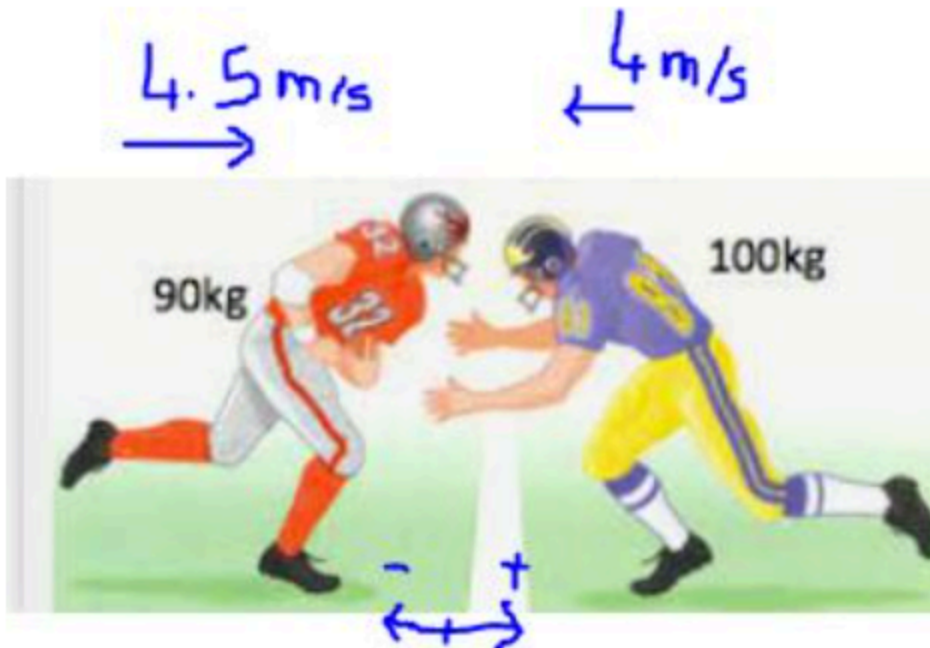
tomatoe



bouncing ball



### Question 8 10 pts



They collide and stick. They will move @ right or left or stop?

refer to previous lecture and video we watched.

hint: the one with the larger MOMENTUM wins



@right

☐

@left

☐

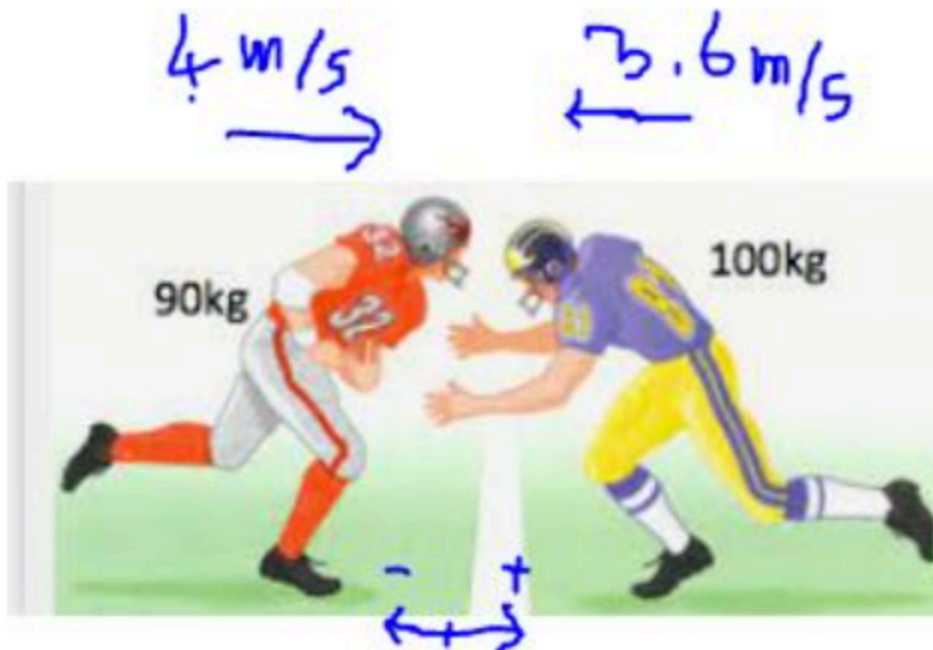
they stop

☐

not enough information

☐

Question 9 10 pts



They collide and stick.

What happens next?

If no one win then? what?

☐

not enough info

☐

they move @ right

☐

they move @ left

☐

they stop

Not saved

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